

SOFTWARE ENGINEERING
Lab 1: Requirements Engineering & UML Use-Case Modelling
Problem Statement #22 – Smart Parking Space Rental & License Validator

Name: Dhanya S Prasad

SRN: PES1UG24CS149

Section: C

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record vehicle entry time via license plate scan and compute total billing amount upon exit based on tiered hourly duration.	High	Pass: Accurate fee calculation based on elapsed duration. Fail: Incorrect billing tier applied.	Core billing functionality required for accurate revenue collection.
FR-002	Functional	The system shall allow a vehicle owner to reserve a parking bay in advance through the mobile app by selecting a date, time slot, and duration.	High	Pass: When a vehicle owner submits a reservation request, the system confirms the booking and displays a unique reservation ID within 5 seconds. Fail: No confirmation or ID is generated.	Guarantees bay availability and reduces search time for drivers.
FR-003	Functional	The system shall validate the scanned license plate number against active reservation records before granting gate entry.	High	Pass: A vehicle with a valid reservation is granted entry within 3 seconds of the scan. Fail: An unreserved or mismatched vehicle is granted entry.	Prevents unauthorized vehicles from occupying pre-booked bays.
FR-004	Functional	The system shall notify the parking warden's dashboard when a vehicle remains parked beyond its paid or reserved duration.	Medium	Pass: An overstay alert appears on the warden dashboard within 1 minute of the reserved time lapsing. Fail: No alert is generated after the grace period.	Supports enforcement of parking rules and timely bay turnover.
FR-005	Functional	The system shall allow a vehicle owner to pay the computed parking fee using at least two payment methods (credit/debit card and UPI/mobile wallet).	High	Pass: Payment is confirmed and a digital receipt is generated for both supported methods. Fail: Payment cannot be completed through either method.	Provides flexible and convenient payment options for users.
NFR-001	Nonfunctional (Performance & Security)	The gate camera license plate lookup API shall verify parking reservation validity in under 300 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Lookup exceeds 300 ms or fails security checks.	Ensures smooth traffic flow at entry/exit gates without long queues.
NFR-002	Nonfunctional (Reliability / Availability)	The parking management system shall maintain at least 99.5% uptime during operating hours (6:00 AM to 12:00 AM).	Medium	Pass: Monthly uptime logs show availability of 99.5% or higher. Fail: Logged downtime exceeds the allowed threshold.	Minimizes revenue loss and customer inconvenience from outages.